

Drought Incidence, Humidity Dynamics, and Chikungunya in Brazilian Municipalities

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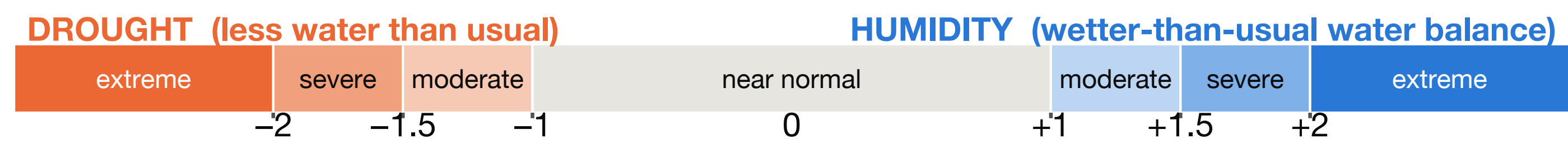
Background

Chikungunya is an increasing public health concern in the Americas. Because *Aedes* mosquitoes require standing water for larval development, **drought and humidity influence transmission**. However, the **time scale** at which they affect disease risk has received limited attention (Morin et al., 2013).

We measure both with the **SPEI** (Standardised Precipitation-Evapotranspiration Index), which captures drought and humidity as anomalies of the water balance at multiple time scales, from short-term rainfall variability (**SPEI-1**) to longer climatic regimes (**SPEI-12**) (Vicente-Serrano et al., 2010). “Humidity” here denotes a wetter-than-usual water balance ($\text{SPEI} > 0$), not air humidity, which enters the models only as a control.

How to read the SPEI

0 = usual water balance for that place and season; below -1 = drought, above $+1$ = humidity



Time window over which the water balance is measured:



Methods

Data. Quarterly data for all **5,556 Brazilian municipalities, 2015–2020**; the dataset contains 122,016 municipality-quarters with complete lagged climate data. Confirmed chikungunya cases from the national surveillance system (SINAN). Gridded daily climate aggregated to municipal quarters; SPEI computed at 1, 3, 6 and 12 months (Begueria et al., 2014).

Design. We compare **each municipality with itself over time**: quarters drier or more humid than that municipality’s usual are compared with its other quarters, after removing nationwide seasonal and epidemic waves (municipality and quarter fixed effects). Drought and humidity are measured in the **previous quarter**, so they precede the cases; temperature, air humidity and daily temperature range are held constant; population is the exposure.

Model. Case counts are analysed with a Poisson regression suited to outbreaks (pseudo-maximum likelihood; Santos Silva & Tenreiro, 2006), with uncertainty allowing for correlation within municipalities:

$$\mathbb{E}[\text{Cases}_{it}] = \text{pop}_{it} \cdot \exp\left(\sum_k \beta_k \text{SPEI}_{i,t-1}^{(k)} + \gamma' X_{i,t-1} + \alpha_i + \lambda_t\right), \quad k \in \{1, 3, 6, 12\}$$

i.e. cases = population $\times$ risk, where i is the municipality, t the quarter, k the SPEI scale, X the climate controls, α_i and λ_t the fixed effects.

What we test. (i) each window on its own; (ii) whether the effect bends (curved relationship); (iii) whether it changes with temperature; (iv) all four windows together; (v) robustness: adjusting for the Premises Infestation Index.

We ask

Do **drought and humidity** at different time scales relate to chikungunya through **distinct mechanisms**?
Is the relationship **non-linear** — can heavy rainfall reduce transmission by **flushing larvae** away (Benedum et al., 2018)?
Does **temperature** modify these effects?

Results

+53% cases

per unit increase in short-term humidity
(SPEI-1, IRR 1.53, $p = 0.016$)

–24% cases

per unit increase in long-term humidity
(SPEI-12, all four scales together, $p = 0.041$)

Inverted U

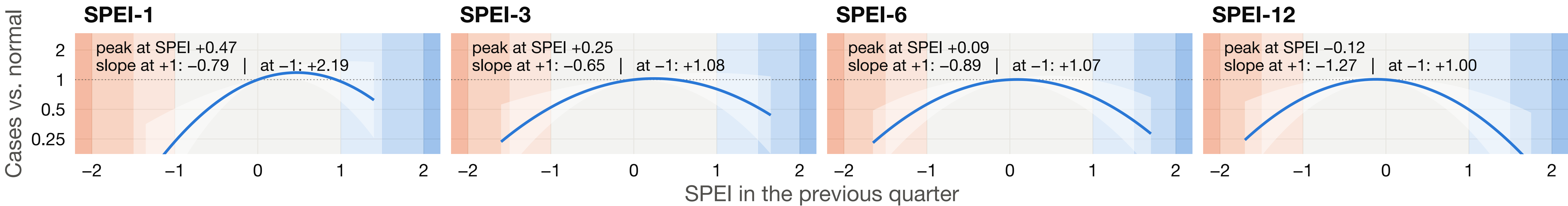
at every time scale: cases fall under extreme
humidity (quadratic term $p < 0.01$)

How to read these results

- **One SPEI unit** = one standard deviation of the local water balance: from *near normal* to *moderate humidity* on the scale above.
- **Each scale on its own**: short-term humidity (SPEI-1) has the strongest association (IRR 1.53); SPEI-3 and SPEI-6 are weaker and not significant; SPEI-12 is negative (IRR 0.86) but not significant alone.
- **All four scales together**: SPEI-1 stays positive ($\beta = 0.64$, $p = 0.002$) and SPEI-12 becomes significantly negative ($\beta = -0.28$, $p = 0.041$); the four scales are jointly significant (Wald $p < 0.001$).
- Results were similar after adjusting for the Premises Infestation Index.

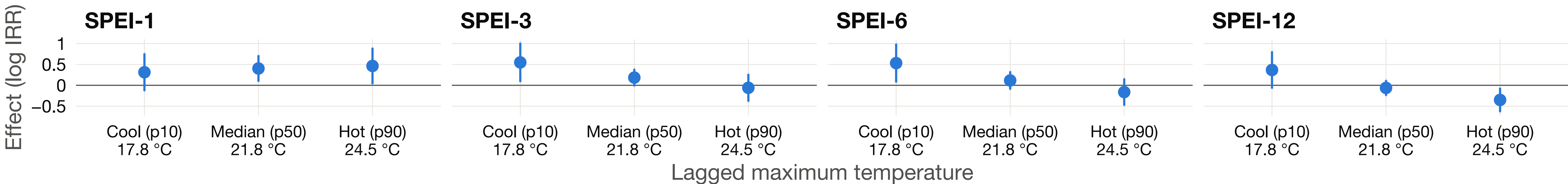
Inverted-U relationship at every time scale: incidence falls under extreme humidity

Quadratic models (curvature $p < 0.01$ at all scales); ribbon = 95% CI; curves drawn within the central 98% of observed SPEI; slopes = log change per SPEI unit at -1 / $+1$



Temperature modifies the effect of longer time scales, not of short-term humidity

SPEI $\times$ temperature interaction: SPEI-6 $p = 0.036$, SPEI-12 $p = 0.026$, SPEI-1 $p = 0.638$ (effect of SPEI shown at cool, median and hot quarters)



Conclusions

- **Short- and long-term drought and humidity are associated with chikungunya in opposite directions.** Recent rainfall appears to increase transmission by creating breeding sites, whereas sustained humidity at longer time scales may reflect environmental conditions that reduce vulnerability.
- **The inverted-U relationship** is consistent with a potential larval-flushing mechanism under extreme wet conditions.
- **Temperature** modifies the effect of longer time scales (SPEI-6, SPEI-12) but not of short-term humidity (SPEI-1).
- **Multi-scale climate indicators may improve early warning systems** for arboviral diseases, because single-scale analyses can mask opposing temporal dynamics.

References

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Data: SINAN (Ministry of Health, Brazil); gridded climate data; IBGE population.